

PEDRAM MOHAMMADI

☎ (604)910-1109 | ✉ pedram.mohammadi.eng@gmail.com | in PedramMohammadi | 🌐 pedrammohammadi.github.io

PROFESSIONAL SUMMARY

- Innovative and results-driven expert in the development of image and video processing algorithms, with a specialization in visual quality enhancement across SDR/HDR formats, tone mapping, color processing, and video encoding. Demonstrated success in delivering patented, production-ready solutions across academic and commercial domains. Proven ability to translate perceptual principles into hardware-efficient designs using Python, MATLAB, C/C++, OpenCV, and FFMPEG.

EDUCATION

Ph.D. in Electrical and Computer Engineering

2015 - 2020

University of British Columbia, Canada

- Thesis: High Dynamic Range (HDR) video processing
- Awards: Full graduate scholarship, International Tuition Award

M.A.Sc. in Electrical Engineering

2012 - 2014

Ferdowsi University of Mashhad, Iran

- Thesis: Subjective and Objective Image and Video Quality Assessment
- Published a widely cited survey paper (300+ citations) on image/video quality evaluation

PROFESSIONAL EXPERIENCE

Senior Video Quality Engineer

2021 - Present

NETINT Technologies Inc., Canada

- Developed multiple patented rate control algorithms (frame-level, block-level, two-pass, constant rate factor) improving BD-rate metrics (PSNR, SSIM, VMAF) by 5 - 15% across VoD and real-time scenarios
- Designed and optimized SDR $\longleftrightarrow$ HDR tone mapping suite, outperforming state-of-the-art by 5 - 8% in objective quality metrics, integrated directly into hardware
- Partnered closely with RTL/hardware teams to translate algorithms into hardware-efficient implementations, including fixed-point adaptations and logic simplifications
- Led end-to-end pipeline testing and tuning across VPU blocks, maintaining visual quality standards, identifying regressions, and coordinating improvements with cross-functional teams
- Acted as internal quality gatekeeper, providing feedback and tuning strategies to feature owners to ensure perceptual excellence in final video output

Scientific R&D Consultant

2020 - 2021

KPMG LLP, Canada

- Evaluated clients' advanced R&D efforts in algorithm design, image/video processing, and related technologies to determine CRA R&D tax relief eligibility
- Helped clients maximize R&D tax credits, achieving 20 - 30% increases (totaling \$500k - \$2M), enabling continued investment in innovation
- Managed concurrent client engagements under tight deadlines, developing technical reports and justifications aligned with both financial and scientific rigor

Application Developer (Video Processing)

2018 - 2019

TELUS Communications Inc., Canada

- Transitioned PhD research into a real-time SDR-to-HDR video conversion system used in content delivery pipeline for 1080p broadcast
- Collaborated with CTO, directors, and engineers to define visual quality goals and meet real-time constraints, verified through objective and subjective testing
- Delivered successful technology transfer and product deployment impacting subscriber-facing video quality

Research Scientist (HDR Imaging & Quality Enhancement)

2015 - 2020

The University of British Columbia, Canada

- Conducted PhD research on perceptual SDR-to-HDR video enhancement, modeling contrast/brightness trade-offs and color fidelity in the perceptual domain
- Developed adaptive segmentation, tone mapping, color adjustment and flicker mitigation algorithms using visual system models and information theory
- Conducted extensive subjective studies and applied objective quality metrics (HDR-VDP, PU-SSIM) to validate 70 - 80% improvements over state-of-the-art
- Work led to multiple IEEE publications and a patented conversion pipeline now licensed in industry

PATENTS

- **P. Mohammadi**, J. Li, E. Andrade Neto, and J. Lou "Methods and Apparatus for Video Rate Control," Patent filed, filing date: January 2024.
- **P. Mohammadi**, E. Andrade Neto, and J. Lou "Image processing methods, devices, electronic devices and storage media," China Patent Office, Application No. CN116167950B, Publication date: April 2023.
- P. Nasiopoulos, S. Ploumis, M. T. Pourazad, R. Boitar, and **P. Mohammadi**, "Methods and Apparatuses for Tone Mapping and Inverse Tone Mapping," U.S. Patent, Application No. US11100888B2, Publication date: January 2019.

PUBLICATIONS

For a complete list of my publications please click [here](#)

WORK AUTHORIZATION

Currently holding a Canadian citizenship